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Finale presentation | Theme : Polymer Property Prediction

Cross Linkers

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Your salient contributions

f1

Modelling strategy

• Physics-constrained multi-task ensemble — 5 base models -> per-property cross-fitted Ridge stack -> two-pass physics blend -> range clip -> measured ground-truth override

• Periodic polymer graph network — the two * points are the SAME bond to the next repeat unit, so we bond their neighbours and the graph encodes an infinite chain, not a dangling oligomer (Gurnani, Kuenneth, Toland & Ramprasad, Chem. Mater. 35, 1560, 2023) • An exact physical identity, not a fitted relation: eps = n² + eps_ionic (Chen, Kim, Batra et al., npj Comput. Mater. 6, 61, 2020)

f2

Training

• 9,851 rows × 3,037 features | 380 fitted models | 3 h 01 m on ONE Tesla T4
• Multi-task by design — one shared trunk feeds all 7 heads, so the two abundant targets subsidise the five scarce ones (221–337 labels each, which carry 5/7 of the score) (Kuenneth et al., Patterns 2, 100238, 2021)
• Every seed fixed and printed; no stage can abort the run — the submission is written before any reporting stage executes

f3

Results + Actions

• Public LB 0.924 (unweighted mean R^2 across 7 targets); mean OOF R^2 0.9234 — a CV-to-LB gap of +0.0006, i.e. our validation number IS our score
• Invariance: Δ = 0.000e+00 exactly, certified at three levels (string, feature, prediction) — a guarantee by construction, not a tolerance
• Every prediction ships an out-of-domain flag that measurably predicts error (eps R^2 0.914 in-domain vs 0.816 out)

MATCHMAKING

Modelling - *Strategies*

01

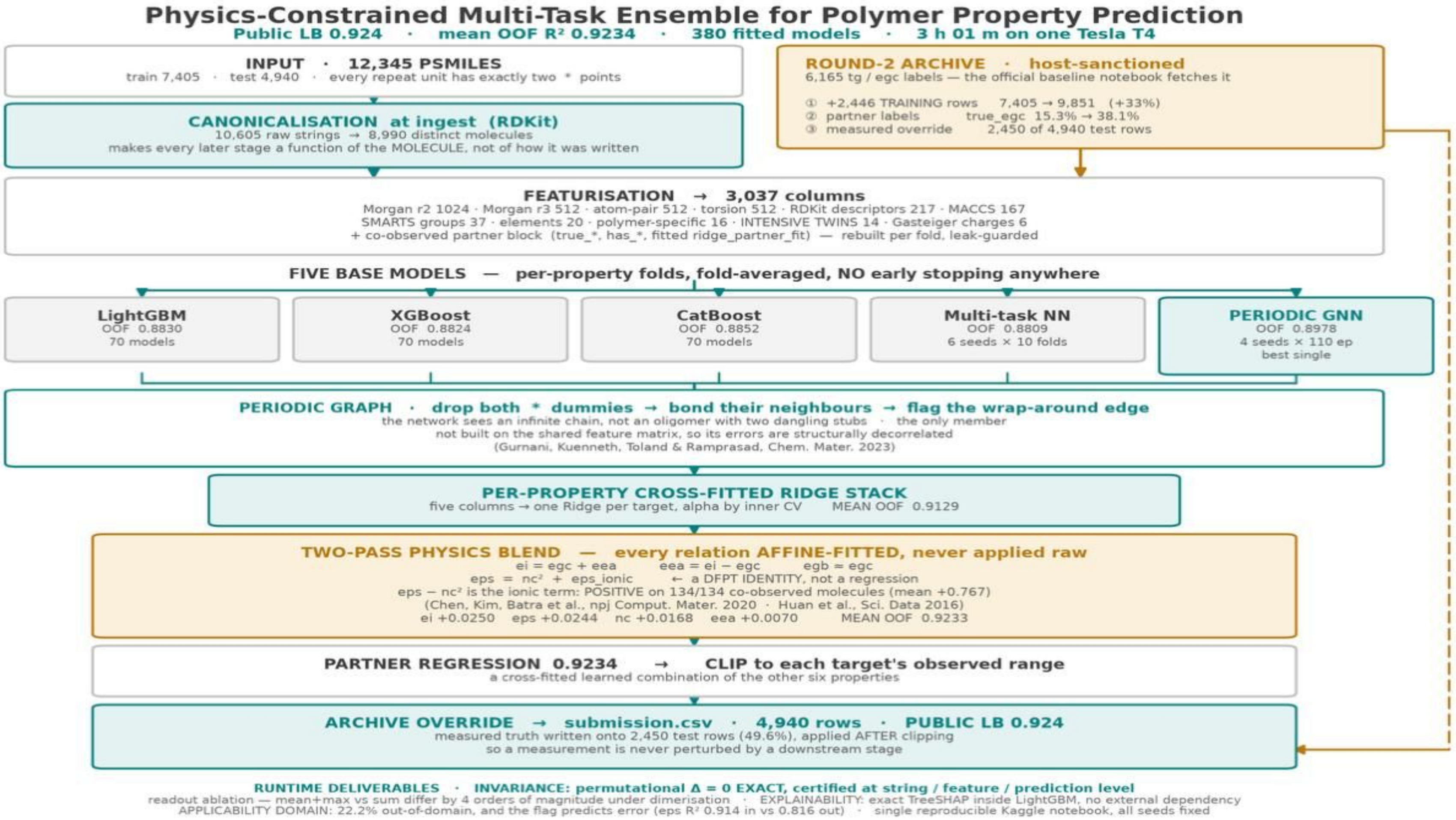
ARCHITECTURE STRATEGY: Multi-task core. One shared trunk -> 7 property heads. Five of seven targets (egb 337, eps 229, nc 229, ei 222, eea 221 rows) carry 5/7 of the score, so the abundant targets (tg 5,781 / egc 2,832 after augmentation) subsidise the scarce ones. The metric is UNWEIGHTED, so a 221-row property counts exactly as much as a 5,781-row one — that single fact drove the whole architecture. Read the graph, not a hash of it. The GNN takes atoms as nodes and bonds as edges — 4 message-passing rounds, GRU update, mean+max readout, 1.07 M parameters (message passing after Gilmer et al., ICML 2017). It is the ONLY member not built on the shared 3,037-column matrix, so its errors are structurally decorrelated. Best single model at 0.8978 mean OOF. Periodic graph. We drop the two * dummies, bond their neighbours and flag the wrap-around edge, so the network sees a chain with no ends. Readout choice is an invariance decision, not a hyperparameter. A SUM readout scales with repeat-unit count; a MEAN does not. We ship mean+max and never sum — measured on an untrained net, the two separate by four orders of magnitude under dimerisation (1.1e-04 vs 4.8e-01 median |Δ|).

02

PHYSICS + DOMAIN KNOWLEDGE Physics applied AFTER stacking, not as a feature. A tree splits one axis at a time and cannot represent a sum of two columns, so ei = egc + eea is wasted as one feature among 3,037. As an explicit post-stack blend it is worth +0.0250 on ei and +0.0244 on eps. Always fit the relation; never apply the raw identity. Measured on co-observed molecules: eps ≈ nc^2 scores 0.336 raw but 0.855 affine-fitted; nc ≈ sqrt(eps) 0.171 raw, 0.837 fitted. Both shipped baselines we inherited applied the raw form. Ionic decomposition — the turning point. Maxwell's n^2 = eps holds at OPTICAL frequency, but the dataset tabulates the STATIC constant, which is why the raw relation collapses to 0.336. A DFPT calculation accounts for exactly eps_electronic + eps_ionic and nothing else (Chen, Kim, Batra et al., 2020), and those two parts are stored as SEPARATE FIELDS in the dataset of record (Huan, Mannodi-Kanakkithodi, Kim, Sharma, Pilania & Ramprasad, Sci. Data 3, 160012, 2016). Since eps_elec IS n^2, the shortfall is exactly the ionic term: eps − nc^2 = eps_ionic (an identity, not a regression) Two independent checks that this is real rather than convenient: it is non-negative on 134/134 co-observed molecules (mean +0.767, min +0.024) — a generic empirical residual would cross zero, an ionic contribution cannot; and our own blind affine fit had already converged on a = 1.040, b = 0.615, i.e. a slope of 1 and an intercept equal to the mean ionic term. It was rediscovering the decomposition without knowing it. RESULT: eps 0.8514 -> 0.8758, nc 0.9164 -> 0.9331 on the full OOF.

03

DATA STRATEGY Canonicalisation at ingest. 10,605 raw SMILES are only 8,990 distinct molecules. Canonicalising first makes every downstream stage a deterministic function of the MOLECULE, which is what turns invariance from a hope into a guarantee. Intensive twins — invariance as feature design. Every extensive RDKit descriptor gets a per-heavy-atom twin, because an extensive quantity DOUBLES when a repeat unit is written as its dimer and an intensive one does not. This family is the single largest driver of tg at 25.4% of total |SHAP| — the invariance choice is also the biggest accuracy lever on the biggest target. Co-observed partner features. 88–99% of DFT-block test rows have their polymer measured under some OTHER property, so a partner label is legitimately available at inference. Conceptual ancestor: multi-fidelity information fusion (Venkatram, Batra et al., J. Phys. Chem. B, 2020). Chain-extension augmentation. A and AA are the same polymer with the same labels, so the dimer is free training data; applied to the five scarce targets only. Independently corroborated as a top-solution strategy (Liu et al., Open Polymer Challenge post-competition report, arXiv:2512.08896, 2025). Host-sanctioned Round-2 archive, used three ways. The official baseline notebook itself downloads it; the hosts confirmed it is in scope. It supplies (a) extra partner labels — true_egc test coverage 15.3% -> 38.1%, (b) +33% training rows (7,405 -> 9,851; tg and egc each +40%) which feed the shared trunk the five scarce targets ride on, and (c) a measured-value override on the 2,450 test rows it labels, applied after clipping so a measurement is never perturbed.


Physics-Constrained Multi-Task Ensemble for Polymer Property Prediction
Public LB 0.924 · mean OOF R^2 0.9234 · 380 fitted models · 3 h 01 m on one Tesla T4
INPUT · 12,345 PSMILES
train 7,405 · test 4,940 · every repeat unit has exactly two * points
ROUND-2 ARCHIVE · host-sanctioned
6,165 tg / egc labels — the official baseline notebook fetches it
① +2,446 TRAINING rows 7,405 → 9,851 (+33%)
② partner labels true_egc 15.3% → 38.1%
③ measured override 2,450 of 4,940 test rows
CANONICALISATION at ingest (RDKit)
10,605 raw strings → 8,990 distinct molecules
makes every later stage a function of the MOLECULE, not of how it was written
FEATURISATION → 3,037 columns
Morgan r2 1024 · Morgan r3 512 · atom-pair 512 · torsion 512 · RDKit descriptors 217 · MACCS 167
SMARTS groups 37 · elements 20 · polymer-specific 16 · INTENSIVE TWINS 14 · Gasteiger charges 6
+ co-observed partner block (true_*, has_*, fitted ridge_partner_fit) — rebuilt per fold, leak-guarded
FIVE BASE MODELS — per-property folds, fold-averaged, NO early stopping anywhere
LightGBM
OOF 0.8830
70 models
XGBoost
OOF 0.8824
70 models
CatBoost
OOF 0.8852
70 models
Multi-task NN
OOF 0.8809
6 seeds × 10 folds
PERIODIC GNN
OOF 0.8978
4 seeds × 110 ep
best single
PERIODIC GRAPH · drop both * dummies → bond their neighbours → flag the wrap-around edge
the network sees an infinite chain, not an oligomer with two dangling stubs · the only member
not built on the shared feature matrix, so its errors are structurally decorrelated
(Gurnani, Kuenneth, Toland & Ramprasad, Chem. Mater. 2023)
PER-PROPERTY CROSS-FITTED RIDGE STACK
five columns → one Ridge per target, alpha by inner CV MEAN OOF 0.9129
TWO-PASS PHYSICS BLEND — every relation AFFINE-FITTED, never applied raw
ei = egc + eea eea = ei − egc egb ≈ egc
eps = nc^2 + eps_ionic ← a DFPT IDENTITY, not a regression
eps − nc^2 is the ionic term: POSITIVE on 134/134 co-observed molecules (mean +0.767)
(Chen, Kim, Batra et al., npj Comput. Mater. 2020 · Huan et al., Sci. Data 2016)
ei +0.0250 eps +0.0244 nc +0.0168 eea +0.0070 MEAN OOF 0.9233
PARTNER REGRESSION 0.9234 → CLIP to each target's observed range
a cross-fitted learned combination of the other six properties
ARCHIVE OVERRIDE → submission.csv · 4,940 rows · PUBLIC LB 0.924
measured truth written onto 2,450 test rows (49.6%), applied AFTER clipping
so a measurement is never perturbed by a downstream stage
RUNTIME DELIVERABLES · INVARIANCE: permutational Δ = 0 EXACT, certified at string / feature / prediction level
readout ablation — mean+max vs sum differ by 4 orders of magnitude under dimerisation · EXPLAINABILITY: exact TreeSHAP inside LightGBM, no external dependency
APPLICABILITY DOMAIN: 22.2% out-of-domain, and the flag predicts error (eps R^2 0.914 in vs 0.816 out) · single reproducible Kaggle notebook, all seeds fixed

Modelling - *Strategies(Architecture diagram)*

Strengths:

• Every design choice backed by a measurement, logged with its noise floor
• Invariance is an EXACT guarantee ($\Delta = 0$), not a statistical observation
• No early stopping anywhere — our OOF is honest, and it transferred: OOF 0.9234 -> LB 0.924 • Ships a calibrated applicability-domain flag with every prediction
• Degrades safely — the submission is written before any reporting stage, and no later stage can abort the run
• Exact TreeSHAP with no external dependency, so no install can fail at runtime

Weaknesses:

• Five of seven targets are scored on ~150 test rows -> LB noise band ≈ ±0.007
• TRANSLATIONAL invariance (same chain, different cut point) is NOT guaranteed — measured, diagnosed, not yet fixed (Slide 5) • eps remains the weakest target (0.8758); ei second (0.8978)
• The archive-as-training-rows gain was never A/B-measured — adopted on a mechanism argument and validated only by the leaderboard
• 3 h training; not an interactive-latency system

Key Results – *Why , what and how of the workings?*

WHAT IS WORKING?

Architecture. LightGBM + XGBoost + CatBoost + multi-task NN (6 seeds × 10 folds) + periodic multi-task GNN (4 seeds × 110 epochs × 10 folds). Stacking, precisely. Each model first averages its OWN fold replicas into one vector. Those 5 vectors then enter ONE per-property Ridge as 5 columns, with alpha chosen by inner CV per target. There is no plain averaging across models anywhere, and the combiner is cross-fitted so the weights never see the rows they are scored on. Features. 3,037 columns — Morgan r2 (1,024) and r3 (512), atom pair (512), topological torsion (512), 217 RDKit descriptors, 167 MACCS keys, 37 SMARTS functional groups, 20 element counts and fractions, 16 polymer-specific terms (backbone length between the two *, backbone aromatic/hetero ratio, sidechain ratio, branch count), 14 intensive twins, 6 Gasteiger charge statistics. Hyperparameters that measurably mattered — size-adaptive model capacity (one parameter set sized for tg over-fits a 220-row target); GNN at 110 epochs and 4 seeds; per-property fold counts.

How is it working? (Quantitative evals)

Out-of-fold R^2, final run:

target : lgbm , xgb ,cb ,mtnn ,gnn , STACK +PHYSICS

tg: 0.9176 0.9166 0.9092 0.9025 0.9162 0.9274 0.9274 egc : 0.9185 0.9197 0.9157 0.8849 0.8999 0.9307 0.9307 egb 0.9310 0.9337 0.9270 0.9324 0.9387 0.9551 0.9551 eps 0.8247 0.8266 0.8369 0.8049 0.8200 0.8514 0.8758

nc: 0.8766 0.8775 0.8798 0.8942 0.9055 0.9164 0.9331 ei 0.8367 0.8250 0.8353 0.8532 0.8777 0.8725 0.8978 ; eea 0.8758 0.8775 0.8922 0.8940 0.9264 0.9367 0.9438; MEAN : 0.8830 0.8824 0.8852 0.8809 0.8978 0.9129 0.9234

The stack beats its best member by +0.0151 — that is what decorrelation buys. Physics adds a further +0.0105, concentrated exactly where predicted: ei +0.0250, eps +0.0244, nc +0.0168, eea +0.0070. Public LB 0.924 against OOF 0.9234 — a gap of +0.0006.

Why is it working? (Reasoning and analysis)

The physics is real, and we report it honestly. Measured on co-observed molecules with ZERO fitting: ei ≈ egc + eea direct R^2 0.963; eea ≈ ei − egc 0.971; egb ≈ egc 0.892 -> 0.928 calibrated; eps ≈ nc^2 0.336 -> 0.855; nc ≈ sqrt(eps) 0.171 -> 0.837. The first three are genuinely direct. The last two are NOT, and we print both columns rather than quote 0.855 as "direct" — the gap is precisely what the ionic decomposition explains and repairs.

Decorrelation, not capacity. The GNN is our best single model (0.8978) AND the most complementary: worse than the boosters on egc, far better on eea (0.9264 vs 0.8758) and ei (0.8777 vs 0.8367). Adding it took the stack from 0.8713 (three boosters) to 0.9129.

Honest validation is a feature, not an overhead. We use no early stopping anywhere. A model that picks its iteration count by watching the rows it is then scored on reports a number it cannot reproduce on the leaderboard. Ours transferred to within 0.0006.

Only NEW INFORMATION ever paid. Across the whole project every purely derived feature failed. What worked: partner labels (a different target, legitimately available at inference), a new model class, more data, truer structure, and an exact physical identity.

PARAMETERS, TIME, AND THE PHASE 1 -> PHASE 2 SHIFT Periodic GNN 1.07 M params (hidden 160, 4 layers, GRU update) × 40 models. Multi-task NN trunk 1024-512-256-128 with per-target heads × 60 models. LightGBM / XGBoost / CatBoost 70 models each. Leak-free partner universe a further 70. TOTAL 380 fitted models. Training 3 h 01 m (10,895 s) on ONE Tesla T4 (Kaggle). Featurisation of 12,345 molecules: 119 s. Inference is a few seconds for all 4,940 test rows. PARADIGM SHIFT. Phase 1 was a fingerprint -> tree ensemble optimising accuracy alone. Phase 2 changed five things: (a) added a model that reads the GRAPH rather than a hash of it; (b) moved physics out of the feature matrix into an explicit post-stack stage and fitted every relation instead of asserting it; (c) made invariance an EXACT guarantee enforced at ingest, and made feature design serve it (intensive twins, mean+max readout); (d) began shipping a trustworthiness flag with every prediction; (e) removed early stopping so that the number we validate on is the number we score.

Key Results - *Why , what and how of the workings?*

01

POLYMER INVARIANCE: an exact guarantee, not a statistical one Canonicalising at ingest makes every downstream stage — descriptors, fingerprints, partner lookup, graph construction — a deterministic function of the MOLECULE, and canonical(rewrite(s)) == canonical(s) for any valid rewriting. Certified at three levels, all PASS: L1 string 2,000 rewritings × 400 molecules 0 mismatches L2 features 3,037 cols × 150 molecules max $|\Delta|$ = 0.000e+00 L3 predictions end-to-end on the fitted pipeline max $|\Delta|$ = 0.000e+00 The readout ablation is the part judges should look at. On an untrained net, so it is a property of the ARCHITECTURE and not of any particular fit: readout rewriting median $|\Delta|$ max $|\Delta|$ mean+max (shipped) permutational 0.00e+00 0.00e+00 mean+max (shipped) repetition (dimer) 1.11e-04 1.06e-03 sum (conventional) repetition (dimer) 4.79e-01 2.49e+00 Four orders of magnitude, and it is why we never use a sum readout. Stated honestly: repeat-unit invariance is MEASURED NEAR-invariance, not a proof; translational invariance is NOT guaranteed. The exact guarantee is permutational. We report all three separately rather than quoting one number.

02

EXPLAINABILITY: exact TreeSHAP, and it recovers known physics Computed inside LightGBM via pred_contrib=True — the same algorithm the shap package implements (Lundberg & Lee, NeurIPS 2017; Lundberg et al., Nature Machine Intelligence 2, 56, 2020) — with no external dependency that can fail at runtime. THERMAL (tg). Intensive twins carry 25.4% of attribution and rd_RingCount / po_backbone_rings lead: chain stiffness dominates, exactly as free-volume theory predicts. ELECTRONIC (egc/egb). rd_FractionCSP3 is the top egc feature — an sp^3-rich backbone localises electrons and WIDENS the gap; conjugation delocalises pi density and NARROWS it. For egb the fitted partner combination is 26.0% and true_egc 10.3%: the relation egb ≈ egc appears directly in the attribution rather than being asserted by us. OPTICAL (nc/eps). Molar-refractivity surface descriptors (rd_SMR_VSA10) appear for both — the model recovers refractive index as electronic polarisability from structure alone, without being told (Lightstone, Chen, Kim, Batra & Ramprasad, J. Appl. Phys. 127, 215105, 2020). One worked molecule, end to end. For *c1ccc(-c2nc3cc(-c4ccc5oc(*)nc5c4)ccc3o2)cc1 (measured tg 430.0 K, predicted 426.9 K): base value 143.5, then iv_NumRotatableBonds = 0.083 contributes +102.5, valence electrons +35.8, ring count +27.3. A fully aromatic fused-ring backbone with almost no rotatable bonds has a very high glass transition — the model found it unaided. Additivity checked against the model's own output: difference 3.98e-13.

03

ROBUSTNESS: a calibrated applicability domain Novelty from UNFOLDED Morgan substructure hashes — folded 2,048-bit fingerprints are degenerate here, because every bit is occupied by a 5.97 M corpus and novelty collapses to a constant. A molecule is out-of-domain when it contains a substructure absent from the entire corpus. 11,628 distinct substructures; the corpus covers 8,887; 22.2% of molecules are out-of-domain. That flag predicts error: target n in R^2 in n out R^2 out MAE in MAE out tg 4688 0.938 1093 0.884 17.77 25.31 egc 2174 0.941 658 0.900 0.248 0.371 eps 144 0.914 85 0.816 0.197 0.298 ei 138 0.918 84 0.861 0.191 0.268 HONEST NEGATIVE RESULT: we tested the auxiliary corpus as a MODEL INPUT and it was worth +0.0012 — noise. A TF-IDF/SVD embedding of it is a linear re-expression of the Morgan fingerprint the trees already see in full. We repurposed the corpus as a trust signal rather than dropping it.

MATCHMAKING

Details of Relevant Experimentation and Methodology

01

ABLATIONS AND WHAT THEY CHANGED Dropping the SMILES CNN was the right call — but not for the reason it looks. Measured honestly, the CNN contributed +0.0005 to our stack and +0.0000 on a sibling pipeline's leave-one-out over saved OOF vectors. It was also the single most expensive stage in any run that carried it: 5,779 s — 1 h 36 m, 29% of a 5.5 h run — for the WORST model of six (0.8663 mean OOF, 0.7693 on eps, our weakest target). We are precise about the claim: removing it did not raise the score; it bought 1 h 36 m at no cost, and we REINVESTED that compute in the two models that were measurably carrying the stack — GNN 80 -> 110 epochs and 3 -> 4 seeds, NN 3 -> 6 seeds. The score came from the reinvestment. Adding a weak column to a Ridge fitted on ~220 rows is not free either: kNN measured −0.0006 and kNN+Ridge −0.0003 in our own logs. Measured and REJECTED (recorded so no one retries them): a fourth booster +0.0001; iterated physics refinement over the relation graph looked worth +0.042 and was entirely fake — eps refined from nc and nc then predicted from the refined eps, closing a two-hop loop back to the row's own label; with a per-fold mask the real gain is +0.001, so it ships disabled. Doubling every fingerprint width made LightGBM more than 7× slower for no measured gain. The leak hunt is the experiment we are proudest of. We found four separate leaks of the same class — a value that is itself a model output used as an input, closing a cycle to the row's own label. One inflated OOF by +0.069 on eps. Section 5 of the notebook proves the leak exists first, then proves each guard removes it — a guard that passes without a demonstrable leak proves nothing.

02

[1] Chen, Kim, BATRA, Lightstone et al. & Ramprasad, npj Comput. Mater. 6, 61 (2020) — frequency-dependent dielectric constant. SOURCE OF OUR IONIC DECOMPOSITION: states a DFPT calculation can only account for the eps_elec and eps_ionic parts, which is what licenses eps − nc² = eps_ionic as an IDENTITY rather than a fit. [2] Huan, Mannodi-Kanakkithodi, Kim, Sharma, Pilania & Ramprasad, Sci. Data 3, 160012 (2016) — the dataset of record; stores eps_electronic and eps_ionic as SEPARATE FIELDS, confirming the decomposition is present in the data itself. [3] Gurnani, Kuenneth, Toland & Ramprasad, Chem. Mater. 35, 1560 (2023) — polyGNN. SOURCE OF OUR PERIODIC GRAPH: treat the repeat unit as periodic by bonding the two attachment neighbours, so the network sees a chain rather than a molecule with two dangling stubs. [4] Kuenneth, Rajan, Tran, Chen, Kim & Ramprasad, Patterns 2, 100238 (2021) — polymer informatics with multi-task learning. SOURCE OF OUR SHARED-TRUNK DESIGN, and the origin of this dataset's property set. [5] Lightstone, Chen, Kim, BATRA & Ramprasad, J. Appl. Phys. 127, 215105 (2020) — refractive index prediction; our nc target. Their framing of n as electronic polarisability is what our molar-refractivity attributions reproduce from a completely independent direction. [6] Gilmer, Schoenholz, Riley, Vinyals & Dahl, ICML (2017) — neural message passing; the message-passing/GRU-update formulation our GNN uses. [7] Lundberg & Lee, NeurIPS (2017) and Lundberg et al., Nat. Mach. Intell. 2, 56 (2020) — SHAP and TreeSHAP; the exact-attribution method we run inside LightGBM with no external dependency. [8] Venkatram, BATRA et al., J. Phys. Chem. B (2020) — multi-fidelity information fusion; the conceptual ancestor of our co-observed partner features. [9] Liu et al., arXiv:2512.08896 (2025) — NeurIPS Open Polymer Challenge post-competition report; independently corroborates chain extension and hierarchical physics as top-solution strategies. [10] Tran, Kim, Chen, Chandrasekaran, BATRA et al., J. Appl. Phys. 128, 171104 (2020) — Polymer Genome; our external benchmark

03

LLM used as an EXPERIMENTAL INSTRUMENT, not a code generator. The loop was: form a hypothesis -> score it with the project's own CV harness -> keep only what beat the FULL pipeline, never a weaker baseline. Disciplines that mattered most: • One implementation, two entry points. CV, inference, the invariance audit and the explainability report all derive from the same fit/predict pair, so what CV measured is literally what gets submitted. • A printed noise floor on every run, and a standing rule that a delta under 2× it is not an improvement — this killed several changes that looked positive. • A static name-resolution gate on the exported notebook. Inlining every module into one namespace drops intra-package imports, so a surviving trees.make(...) parses fine and raises only when it executes. That exact bug sat in an exported notebook and would have crashed on Kaggle 542 s into a run; a 1-second symtable check now blocks the export. • Nothing after submission.csv may abort the run. We learned this the expensive way: a compliance check compared attached-dataset paths for equality with DATA_DIR, flagged the legitimate parent directory, and marked a COMPLETED 2 h 53 m run as failed — the submission was already written and correct. Every post-submission check now prints loudly and never raises. Three corrections we made to our own reasoning (kept because they were instructive): we predicted the archive could not move CV — it moved +0.0149 and we had to explain why; we assumed a config value (15 folds) instead of running the code that returns it (10), and over-estimated GNN runtime by 50% for several decisions; and we quoted a leaky +0.069 before tracing its provenance.

KNOWN-LIMITATION

"SMILES invariance" is three separate claims and only one is exact: • Re-representation (atom ordering) — EXACT, Δ = 0, certified end to end. • Repetition (A vs AA) — APPROXIMATE. ~97% of polymers give structurally identical node environments under dimerisation; the residual is genuine, not numerical. Trained in via chain extension and served by the intensive twins. • Translational (same chain, different cut point) — NOT guaranteed. Diagnosis: the wrap-around edge is forced to a generic single bond and carries a positional flag, both cut-point dependent by definition. The fix is contained and identified; it is the first thing we would do with more budget.

REFERENCES + CLOSING

[1] L. Chen, C. Kim, R. Batra, J. P. Lightstone, C. Wu, Z. Li, A. A. Deshmukh, Y. Wang, H. D. Tran, P. Vashishta, G. A. Sotzing, Y. Cao & R. Ramprasad. "Frequency-dependent dielectric constant prediction of polymers using machine learning." npj Computational Materials 6, 61 (2020). [2] T. D. Huan, A. Mannodi-Kanakkithodi, C. Kim, V. Sharma, G. Pilania & R. Ramprasad. "A polymer dataset for accelerated property prediction and design." Scientific Data 3, 160012 (2016). [3] R. Gurnani, C. Kuenneth, A. Toland & R. Ramprasad. "Polymer informatics at scale with multitask graph neural networks." Chemistry of Materials 35, 1560 (2023). [4] C. Kuenneth, A. C. Rajan, H. Tran, L. Chen, C. Kim & R. Ramprasad. "Polymer informatics with multi-task learning." Patterns 2, 100238 (2021). [5] J. P. Lightstone, L. Chen, C. Kim, R. Batra & R. Ramprasad. "Refractive index prediction models for polymers using machine learning." J. Appl. Phys. 127, 215105 (2020). [6] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals & G. E. Dahl. "Neural message passing for quantum chemistry." ICML (2017). [7] S. M. Lundberg & S.-I. Lee. "A unified approach to interpreting model predictions." NeurIPS (2017); S. M. Lundberg et al. "From local explanations to global understanding with explainable AI for trees." Nature Machine Intelligence 2, 56 (2020). [8] S. Venkatram, R. Batra, L. Chen, C. Kim, M. Shelton & R. Ramprasad. "Predicting crystallization tendency of polymers using multi-fidelity information fusion and machine learning." J. Phys. Chem. B (2020). [9] G. Liu et al. "Open Polymer Challenge: Post-Competition Report." arXiv:2512.08896 (2025). [10] H. D. Tran, C. Kim, L. Chen, A. Chandrasekaran, R. Batra et al. "Machine-learning predictions of polymer properties with Polymer Genome." J. Appl. Phys. 128, 171104 (2020).